

THE STRUCTURE OF MATHEMATICAL REALITY

Phase V — Chapter 17

Open Questions and the Shape of the Unknown

Exercises

Self-Study Curriculum

1. Take one of the genuinely open questions from the list. Write a one-paragraph argument for why it might have a definitive answer. Then write a one-paragraph argument for why it might not. What would it take to resolve the question?

2. Choose two fields from the “gaps in personal knowledge” list. For each, state: (a) what question you would bring to that field, (b) what concept from this curriculum connects to it, and (c) what you would expect to gain from studying it.

3. Revisit Exercise 1 from Chapter 1: the two systems that share a structure. Can you now describe that shared structure more precisely? Can you name the field of mathematics where that structure lives? Can you identify the morphisms?
